Round 3 of the conductor-gap study: two-torsion summation polynomials, and the combined descent audited

2026-10-11. research/isogeny_conductor_gap_round3_20261011. Protocol frozen before both leads ran (PROTOCOL.md). Result classes: E4 stage diagnostic (no whole-method or rho comparison); E5 engineering, an audited negative.

Findings

1. **Two-torsion symmetrisation is a quotient isogeny and pays as predicted.** Writing the S_3 system for an H-stable factor base on E/H (Vélu) divides the factor-base polynomial's degree by |H|. At matched base B = 32 or 64 on E, the F4 solving degree falls 1.9–3.7× and F_p row-reduction work 4.1–4.3× (|H| = 2) and 17.6–19.6× (|H| = 4). F4's cost on these systems depends only on that degree.
2. **The crater's richer 2-torsion changes nothing at matched |H|.** On 12 pairs (every D = −7 pair and the $\ell = 2$ pairs of D = −11 and D = −3) the crater has rational E[2] and the floor rank-1 2-torsion; crater E[2] and the floor's and floor model's C4 give the same solving degree and widest matrix, with work within 6 of 34,158 at B = 64.
3. **Round 1′s combined Weil descent is negative in audited sessions.** On the a = 1 Koblitz curve over $F_{2^{17}}$ it verifies in 4 of 4 runs at 324–613 s online (blind 0.9–2.0 s), maximum solving degree 4 (blind 3), 234–529× the blind F4 work paired by run, for at most h = 2 times the relations. The predicted 600–700× fails; round 1′s unaudited 629 and 654× lie above the audited range.

E4: 1,128 F4 systems on 72 curves

24 even-cofactor round-1 prime pairs × crater, floor, floor model; B = 32, 64; H = 1, C2, C4, E[2] where rational; two random and one planted target. Every system completed; F4 agrees with enumeration on all; every planted solution annihilates its basis; the Vélu map passed 32 random-sum checks per curve and subgroup. Run report: reports/run-20261011T1714Z.

base-poly degree	solving degree	widest matrix	mults, random	mults, planted
8	10	45	7,408–7,414	13,903–13,929
16	18	93	34,152–34,158	81,128–81,178
32	34	189	145,630	483,552–483,561
64	66	381	600,510	3,123,348–3,123,354

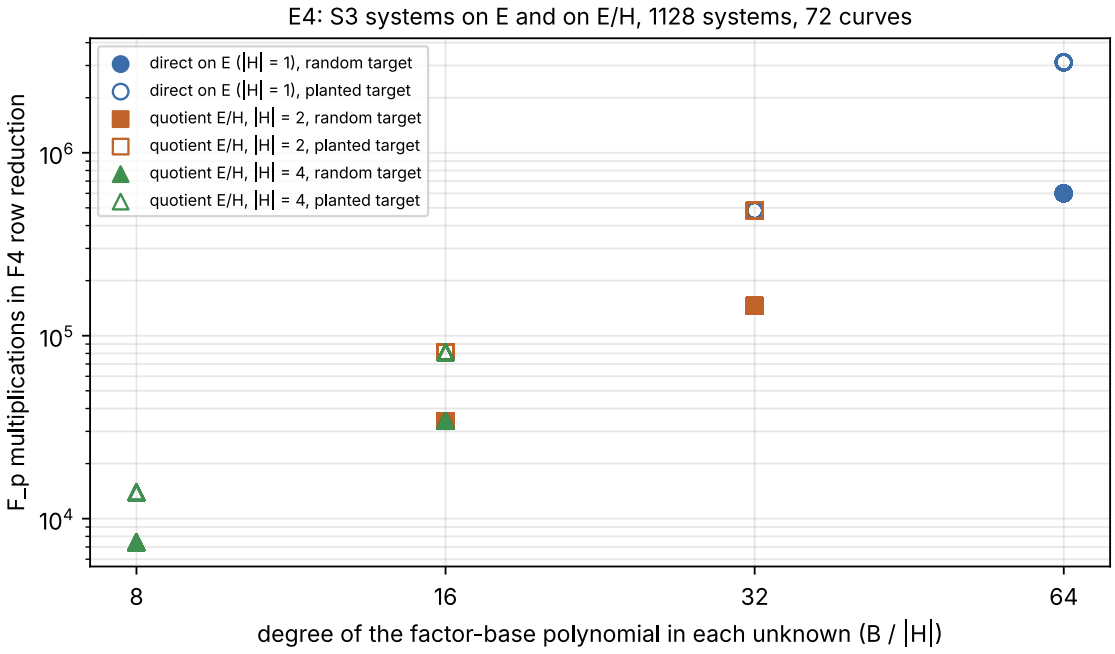


Figure 1: Every E4 system: F_p row-reduction multiplications against the factor-base polynomial degree B/|H|; records e4/records.jsonl.

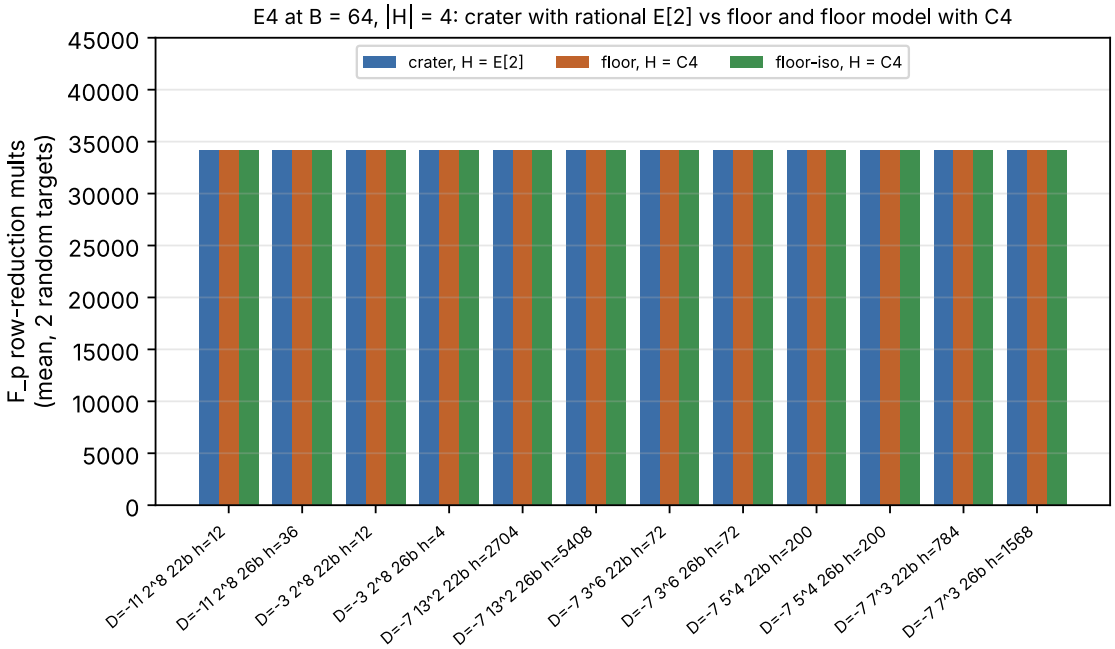


Figure 2: B = 64, |H| = 4: crater E[2] against floor and floor-model C4 on the 12 pairs whose crater has rational E[2].

The protocol expected rational E[2] on the crater only from $\ell = 2$ volcanoes; the run found it on every $D = -7$ crater too (their floors' Frobenius conductors are even). The predictions did not depend on this. Scope: two summands, bivariate systems, split factor-base polynomials; higher m is untested.

E5: two audited ecbench sessions

session, round	blind wall, F4 ops	combined wall, F4 ops	ratio	max degree	peak RSS
t0, 0	0.9 s, 65,072,207	324.5 s, 32,012,560,285	492	4	0.66 GB
t0, 1	1.8 s, 133,408,080	613.2 s, 70,613,284,746	529	4	0.64 GB
t1, 0	2.0 s, 164,172,136	453.5 s, 38,387,092,945	234	4	0.65 GB
t1, 1	1.3 s, 84,877,444	464.5 s, 34,853,985,256	411	4	0.99 GB

Sessions ECBS1h05f404b1f89b and ECBS1he4c2c379c257, 8 records each, all verified, all replays reproduced; isolation earned L1 on 6 and L2 on 2 records each. Solver work depends on each run's algorithm seed: round 1's two identical counts (59,916,380,641) came from two warm-ups with the same seed. S leaves solver XORs unpriced and is a loose lower bound for the combined arm. The h = 6 random-curve timeout of round 1 was not rerun.

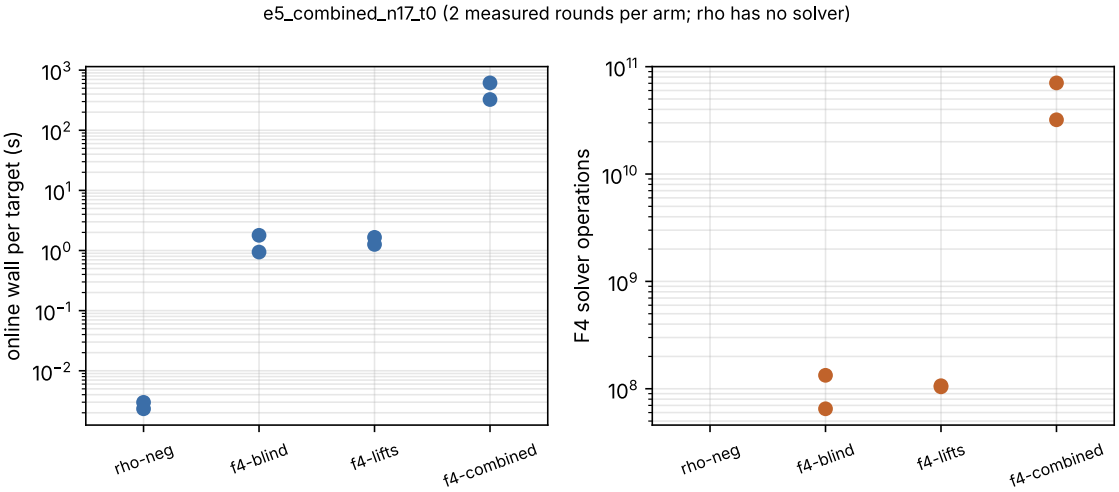


Figure 3: Session ECBS1h05f404b1f89b (target 0, seed 20261010): online wall and F4 operations per arm.

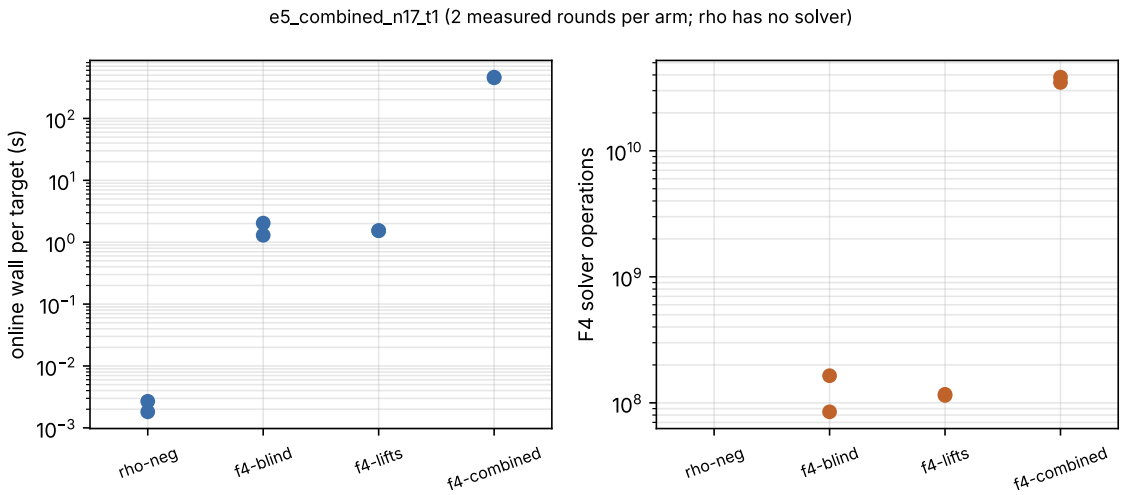


Figure 4: Session ECBS1he4c2c379c257 (target 0, seed 20261011).

Graphs checked

Scoreboard: a round-3 panel added and round 1′s combined-descent row updated to the audited figures. Progress timeline: unchanged (E4 makes no rho comparison; E5′s curve was already plotted by round 1). Built from the round-2 head 75b95656a because main at 1613b90fe does not compile (union-merge damage in koblitz_index_calculus.rs and koblitz_factor_base_search.rs).